

Triplet loss

$$L = \max(0, \underbrace{D(A, P)}_{\text{dist betn Anchor \& Positive}} - \underbrace{D(A, N)}_{\text{dist betn Anchor \& negative}} + m)$$

$m \Rightarrow$ margin

value that encourages a gap betn +ve & -ve distances.

If $D(A, P) + m < D(A, N)$ is satisfied
 $\Rightarrow$ loss is zero (no update needed).

otherwise model adjusts its weights to push apart the negative samples.

If $L > 0$ model needs to adjust so that $D(A, P)$ gets smaller & $D(A, N)$ gets larger.

If $L = 0$ triplet is already well separated so no learning happens.

eg:

Anchor embeddings $A = (1.2, 0.8, 0.5)$

Positive $P = (1.0, 0.9, 0.6)$

Negative $N = (2.5, 1.8, 1.5)$

$$m = 0.5$$

distance metric = euclidean distance.

$$\begin{aligned} \textcircled{1} \quad D(A, P) &= \sqrt{(1.2 - 1.0)^2 + (0.8 - 0.9)^2 + (0.5 - 0.6)^2} \\ &= \sqrt{(0.2)^2 + (-0.1)^2 + (-0.1)^2} \\ &= \sqrt{0.04 + 0.01 + 0.01} \\ &= \sqrt{0.06} \\ &\approx 0.245 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad D(A, N) &= \sqrt{(1.2 - 2.5)^2 + (0.8 - 1.8)^2 + (0.5 - 1.5)^2} \\ &= \sqrt{(-1.3)^2 + (-1.0)^2 + (-1.0)^2} \\ &= \sqrt{1.69 + 1.00 + 1.00} \\ &= \sqrt{3.69} \\ &\approx 1.92 \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad L &= \max(0, D(A, P) - D(A, N) + m) \\ &= \max(0, 0.245 - 1.92 + 0.5) \\ &= \max(0, -1.175) \\ &= 0 \end{aligned}$$

model does not need to update parameters for this triplet because neg is already enough far from anchor.